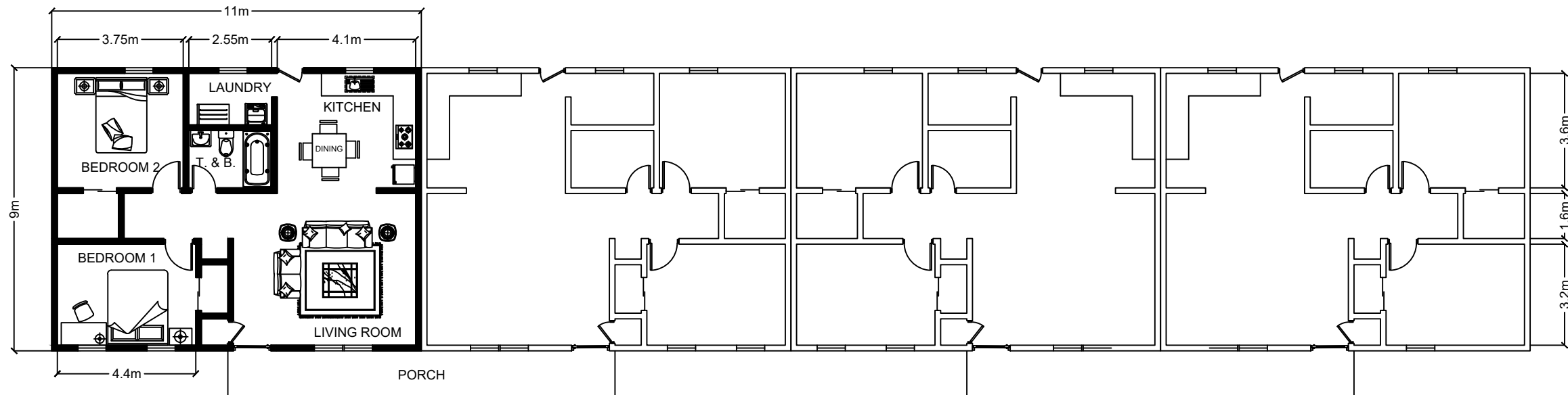
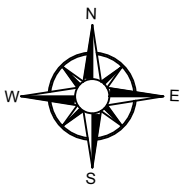
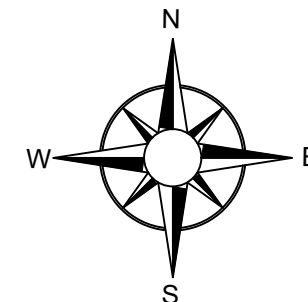
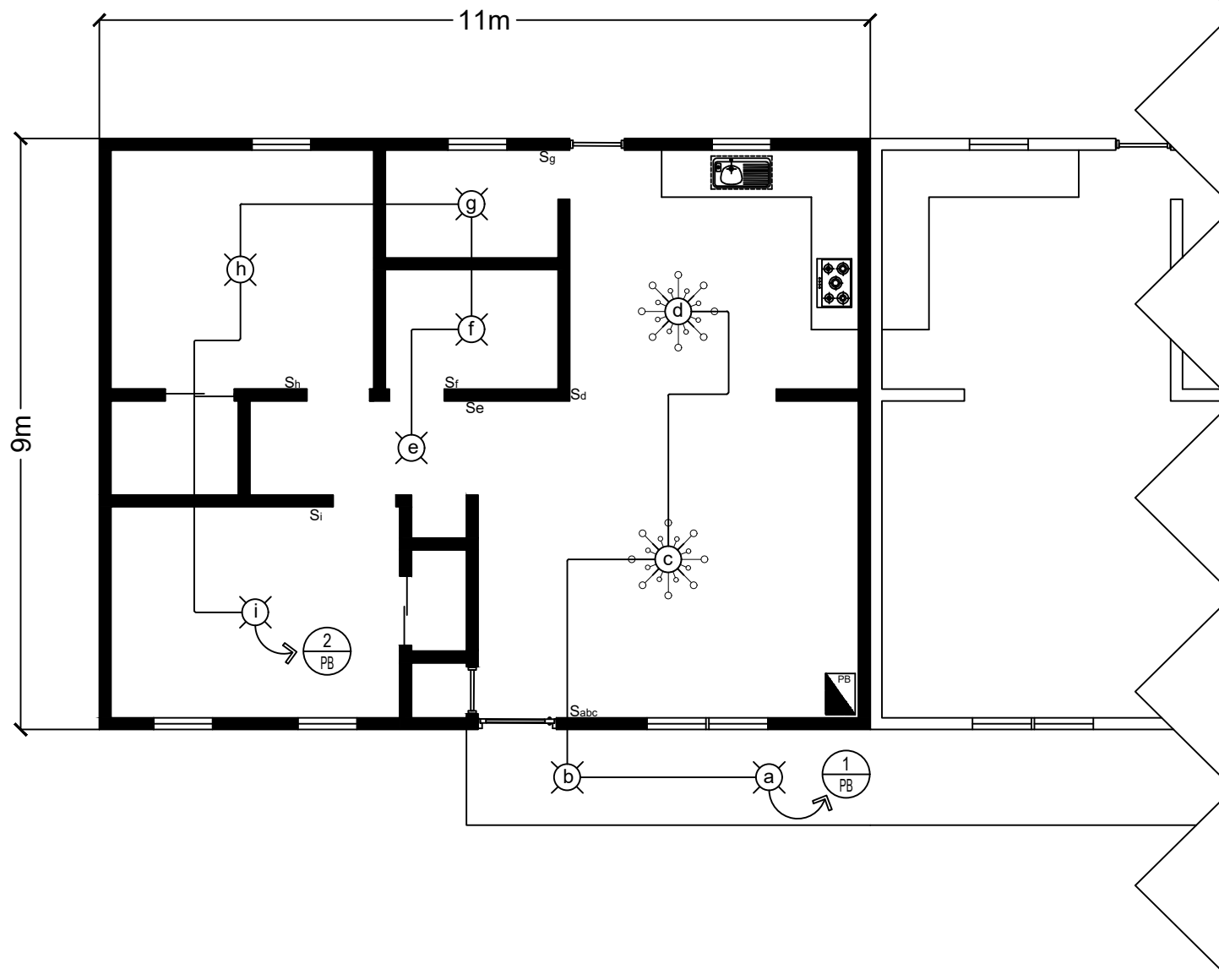




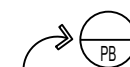
SCHOOL NAME:	COURSE TITLE:	PROJECT TITLE:	SUBMITTED BY:	SUBMITTED TO:	SHEET CONTENT:	GRADE:
 DR. EMILIO B. ESPINOSA SR. MEMORIAL STATE COLLEGE OF AGRICULTURE AND TECHNOLOGY	EE412 - ELECTRICAL SYSTEMS AND ILLUMINATION ENGINEERING DESIGN	MULTI-FAMILY DWELLING FLOOR PLAN	<u>BLADEMIR P. RUBIA</u> BSEE - 4	<u>ENGR. CHRIS JAY D. DIONG-AN</u> INSTRUCTOR	E 1 5	



LEGENDS



Panel Board



Circuit Homerun



Chandelier Lighting Fixture



Ceiling Light



Lighting Circuit Line

Sa

1 Gang Switch

Sabc

3 Gang Switch

SCHOOL NAME:

COURSE TITLE:

PROJECT TITLE:

SUBMITTED BY:

SUBMITTED TO:

SHEET CONTENT:

GRADE:



DR. EMILIO B. ESPINOSA SR.
MEMORIAL STATE COLLEGE
OF AGRICULTURE AND
TECHNOLOGY

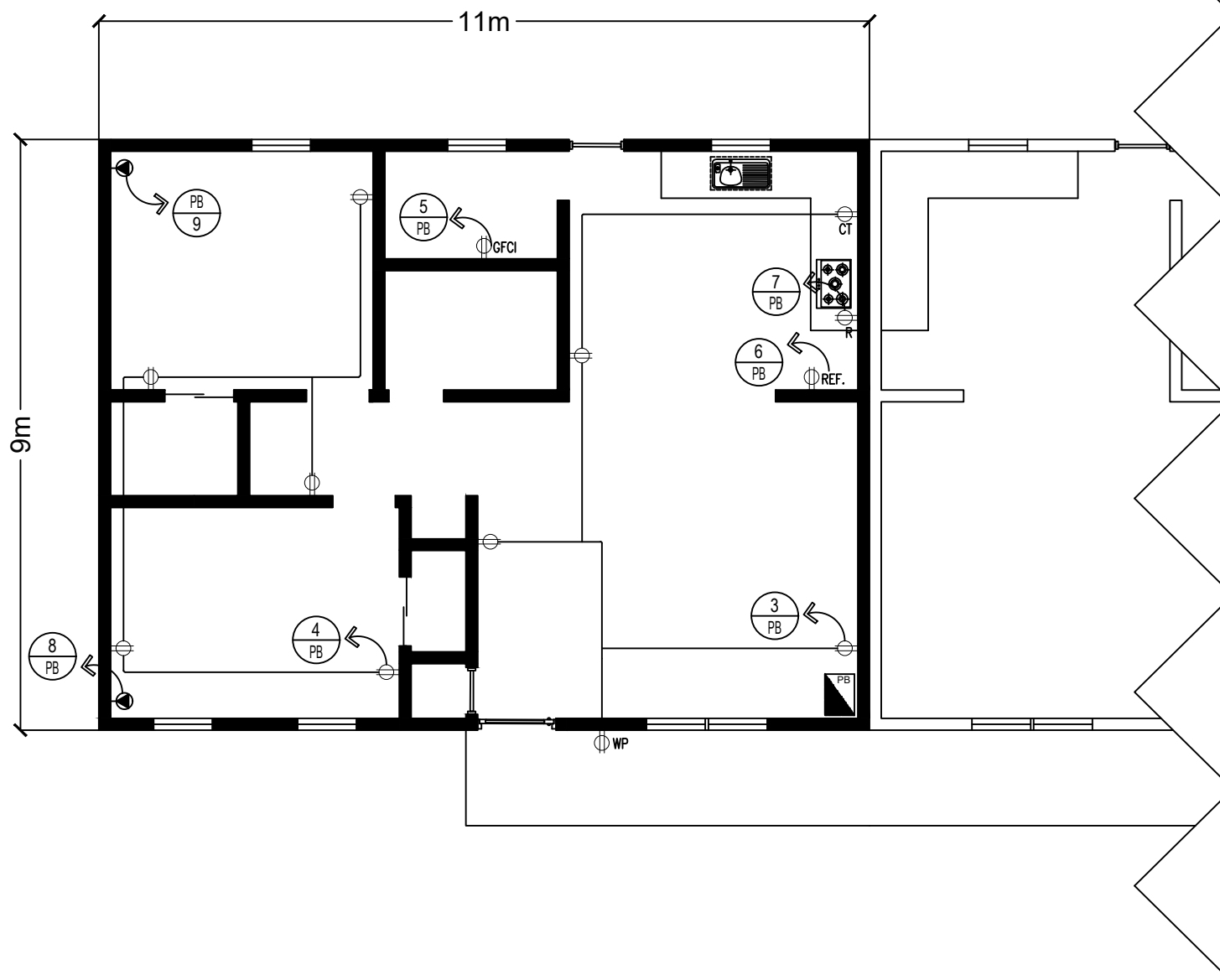
EE412 - ELECTRICAL
SYSTEMS AND ILLUMINATION
ENGINEERING DESIGN

MULTI-FAMILY DWELLING
LIGHTING LAYOUT

BLADEMIR P. RUBIA
BSEE - 4

ENGR. CHRIS JAY D. DIONG-AN
INSTRUCTOR

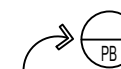
E
2 | 5



LEGENDS



Panel Board



Circuit Homerun



Duplex Convenience Outlet



Range Outlet



Refrigerator Outlet



Counter Top Convenience Outlet

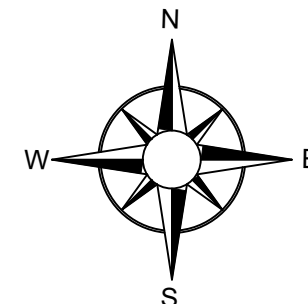


Weather Proof Outlet



Special Purpose Outlet

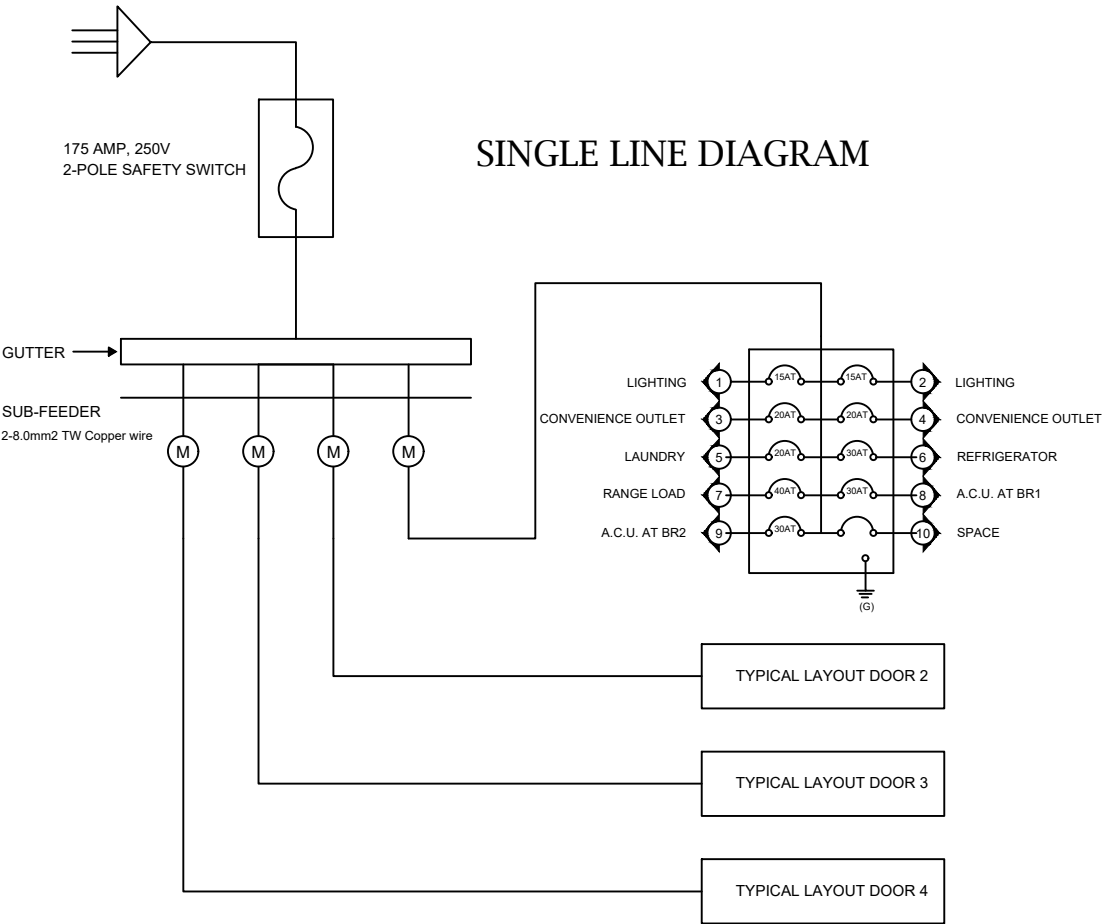
Power Line

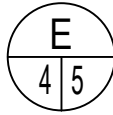


SCHOOL NAME:	COURSE TITLE:	PROJECT TITLE:	SUBMITTED BY:	SUBMITTED TO:	SHEET CONTENT:	GRADE:
 DR. EMILIO B. ESPINOSA SR. MEMORIAL STATE COLLEGE OF AGRICULTURE AND TECHNOLOGY	EE412 - ELECTRICAL SYSTEMS AND ILLUMINATION ENGINEERING DESIGN	MULTI-FAMILY DWELLING POWER LAYOUT	BLADEMIR P. RUBIA BSEE - 4	ENGR. CHRIS JAY D. DIONG-AN INSTRUCTOR	E 3 5	

SCHEDULE OF LOADS

PANEL : "PB"							
CKT. NO.	DESCRIPTION	VOLTS	BREAKER		LOAD IN		SIZE OF WIRE AND CONDUIT
			AT	POLE	VA	AMPS	
1.	4 Ceiling Light @ 100W	230	15	2	400	1.74	1-2.0mm² COPPER WIRE IN 13mmØ CONDUIT
2.	5 Ceiling Light @ 100W	230	15	2	500	2.17	1-2.0mm² COPPER WIRE IN 13mmØ CONDUIT
3.	5- DUPLEX CONVENIENCE OUTLET	230	20	2	1800	7.83	2-3.5mm² COPPER WIRE IN 13mmØ CONDUIT
4.	5- DUPLEX CONVENIENCE OUTLET	230	20	2	1800	7.83	2-3.5mm² COPPER WIRE IN 13mmØ CONDUIT
5.	LAUNDRY	230	20	2	1,500	6.52	2-3.5mm² COPPER WIRE IN 13mmØ CONDUIT
6.	REFRIGERATOR	230	30	2	1,500	6.52	2-3.5mm² COPPER WIRE IN 13mmØ CONDUIT
7.	RANGE	230	40	2	8,000	34.78	2-8.0mm² COPPER WIRE IN 20mmØ CONDUIT
8.	A.C.U. AT BR1	230	30	2	1,119	4.86	2-3.5mm² COPPER WIRE IN 13mmØ CONDUIT
9.	A.C.U. AT BR2	230	30	2	1,119	4.86	2-3.5mm² COPPER WIRE IN 13mmØ CONDUIT
10.	SPACE	230	-	2	-	-	- - -



SCHOOL NAME:	COURSE TITLE:	PROJECT TITLE:	SUBMITTED BY:	SUBMITTED TO:	SHEET CONTENT:	GRADE:
 DR. EMILIO B. ESPINOSA SR. MEMORIAL STATE COLLEGE OF AGRICULTURE AND TECHNOLOGY	EE412 - ELECTRICAL SYSTEMS AND ILLUMINATION ENGINEERING DESIGN	MULTI-FAMILY DWELLING SCHEDULE OF LOADS AND SINGLE LINE DIAGRAM	BLADEMIR P. RUBIA BSEE - 4	ENGR. CHRIS JAY D. DIONG-AN INSTRUCTOR		

BILL OF MATERIALS PER DWELLING UNIT

ITEM	QUANTITY	COST	TOTAL
Single gang Switch	6 set	Php. 128.00	Php. 768.00
3 gang Switch	1 set	Php. 92.00	Php. 92.00
Chandelier Lighting Fixture	2 pieces	Php. 1,500.00	Php. 3,000.00
Ceiling Light	7 pieces	Php. 50.00	Php. 350.00
Lighting Socket	9 pieces	Php. 50.00	Php. 450.00
Duplex Convenience Outlet	9 set	Php. 75.00	Php. 675.00
Counter Top Outlet	1 set	Php. 350.00	Php. 350.00
Range Outlet	1 set	Php. 300.00	Php. 300.00
Refrigerator Outlet	1 set	Php. 200.00	Php. 200.00
GFCI Outlet	1 set	Php. 500.00	Php. 500.00
Weather Proof Outlet	1 set	Php. 350.00	Php. 350.00
A.C.U. Outlet	2 set	Php. 290.00	Php. 580.00
2.0mm² Copper Wire	3 roll	Php. 1,500.00	Php. 4,500.00
3.5mm² Copper Wire	5 roll	Php. 2,200.00	Php. 11,000.00

8.0mm² Copper Wire	50 meters	Php. 80.00	Php. 4,000.00
13mmØ PVC PIPE	25 pieces	Php. 105.00	Php. 2,625.00
20mmØ PVC PIPE	5 pieces	Php. 105.00	Php. 525.00
25mmØ PVC PIPE	5 pieces	Php. 105.00	Php. 525.00
Panel Board	1 set	Php. 1,550.00	Php. 1,550.00
Junction Box	9 pieces	Php. 40.00	Php. 360.00
Utility Box	23 pieces	Php. 20.00	Php. 460.00
Safety Switch (70 amp)	1 piece	Php. 550.00	Php. 550.00
40 Amperes Breaker	1 piece	Php. 400.00	Php. 400.00
30 Amperes Breaker	3 pieces	Php. 110.00	Php. 330.00
20 Amperes Breaker	3 pieces	Php. 110.00	Php. 330.00
15 Amperes Breaker	2 pieces	Php. 110.00	Php. 220.00
Electrical Tape	5 pieces	Php. 60.00	Php. 300.00
GRAND TOTAL			PHP. 35,290.00

SCHOOL NAME:	COURSE TITLE:	PROJECT TITLE:	SUBMITTED BY:	SUBMITTED TO:	SHEET CONTENT:	GRADE:
 DR. EMILIO B. ESPINOSA SR. MEMORIAL STATE COLLEGE OF AGRICULTURE AND TECHNOLOGY	EE412 - ELECTRICAL SYSTEMS AND ILLUMINATION ENGINEERING DESIGN	MULTI-FAMILY DWELLING BILL OF MATERIALS	<u>BLADEMIR P. RUBIA</u> BSEE - 4	<u>ENGR. CHRIS JAY D. DIONG-AN</u> INSTRUCTOR	E 5 5	